- The degree of freedom represents the number of independent observation in the sample.
- The sampling distribution of the test statistic for testing hypothesis about normal population mean is t_{n-1}, when n is small and σ is unknown.
- The sampling distribution of the test statistic for testing equality of two normal population mean is t_{m+n-2} when $m, n < 30$ and the common population variance σ^2 is unknown.
- If $Z \sim N(0,1)$ then $Z^2 \sim \chi^2$ with 1 d.f.
- The uses of χ^2 – distribution are (i) testing the specified variance of a normal population (ii) testing goodness of fit and (iii) testing independence of attributes.
- When expected frequency for a cell is less than 5, it is should be clubbed with the adjacent cells such that the expected frequency in the resultant cell is greater than 5.
- The degrees of freedom for the χ^2 – statistic used for the independence of attributes is $(m-1) \times (n-1)$, where m and n are respectively the number of rows and columns in a contegency table.
- The expected cell frequency testing independence of attributes is calculated as
$$\frac{\text{Row total} \times \text{Column total}}{\text{Sample Size}}$$
- The expected cell frequency in testing goodness of fit is calculated as sample size × {probability for the corresponding cell}

EXERCISE 2

278RKM

I. Choose the best answer.

1. Student's 't' distribution was found by
   a) Karl Pearson b) Laplace
   c) R.A. Fisher d) William S.Gosset
2. Support of student's t random variable is
   a) $-\infty < t \leq 0$ b) $0 \leq t < \infty$
   c) $-\infty < t < \infty$ d) $0 \leq t \leq 1$
3. Paired t-test is applicable when the observations in both the samples are
   a) Paired b) Correlated
   c) Equal in number d) all the above
4. The number of degrees of freedom for the test statistic $t = \frac{(\bar{x}-\mu)}{s/\sqrt{n}}$ is
   a) $n-1$ b) n c) $n-2$ d) $n+1$
5. Standard error of difference between two sample means in the case of small samples is
   a) $\sigma^2\sqrt{\frac{1}{m}+\frac{1}{n}}$ b) $\frac{\sigma_1^2}{m}+\frac{\sigma_2^2}{n}$ c) $s_p\sqrt{\frac{1}{m}+\frac{1}{n}}$ d) $\sqrt{\frac{\sigma_1}{m}+\frac{\sigma_2}{n}}$

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